A Unit-Norm Identity Relating the Product of Pisot Boundary Roots to the Golden Ratio

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Abstract

Let r_n denote the largest real root of $x^n - x - 1$ for $n \geq 2$. The root $r_2 = \varphi$ is the golden ratio, $r_3 = \rho$ is the plastic constant (the smallest Pisot number), and r_4 is the first root in the family that fails the Pisot condition. We prove that the minimal polynomial of $\rho \cdot r_4$ over $\mathbb{Q}$, reduced modulo the golden ratio polynomial $t^2 - t - 1$, yields the remainder $t - 1$ exactly—the fundamental unit φ^{-1} of the ring $\mathbb{Z}[\varphi]$, with algebraic norm -1. For all other products $r_m \cdot r_n$ with $2 \leq m < n \leq 8$, the corresponding norm has absolute value at least 11, growing super-exponentially with $\max(m, n)$. As a corollary, the topological entropies satisfy $\ln r_2 = \ln r_3 + \ln r_4 + O(10^{-4})$, while $|\ln r_m - \ln r_n - \ln r_p| = O(1)$ for all other triples. The $(3, 4)$ pair is thus uniquely distinguished within the $x^n = x + 1$ family by its proximity to the golden ratio, and the proximity has an exact algebraic origin.

1 Introduction

The polynomials $f_n(x) = x^n - x - 1$ for $n \geq 2$ play a distinguished role in the theory of Pisot–Vijayaraghavan numbers. Each f_n is irreducible over $\mathbb{Q}$ [4], and for $n \leq 3$, the largest real root r_n is a Pisot number: an algebraic integer greater than 1 whose conjugates all lie strictly inside the unit disk. For $n = 4$, this property fails—f_4 has a pair of conjugate roots of modulus approximately 1.0606, placing r_4 outside the Pisot class. The transition at $n = 3 \to 4$ is the *Pisot boundary* of the family [1].

The roots $r_2 = \varphi = (1+\sqrt{5})/2$ (the golden ratio), $r_3 = \rho \approx 1.32472$ (the plastic constant, also the smallest Pisot number [5]), and $r_4 \approx 1.22074$ are among the most studied algebraic integers. The numerical observation

$$\rho \cdot r_4 = 1.61714\ldots \approx 1.61803\ldots = \varphi \tag{1}$$

has appeared in recreational mathematics contexts, but to the author's knowledge, no algebraic explanation has been given. The product agrees with φ to 0.055%, which is surprisingly tight given that ρ and r_4 belong to unrelated number fields of degrees 3 and 4 respectively.

In this note, we provide an exact algebraic identity that explains the approximation (1) and prove that it is unique within the family $\{r_n\}$.

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2 The Minimal Polynomial of $\rho \cdot r_4$

Since ρ is a root of $f_3(x) = x^3 - x - 1$ and r_4 is a root of $f_4(x) = x^4 - x - 1$, the minimal polynomial of their product $t = \rho \cdot r_4$ over $\mathbb{Q}$ can be obtained via the resultant

$$P(t) = \mathrm{Res}_x\big(x^3 - x - 1,\ t^4 - tx^3 - x^4\big),$$

where the second polynomial arises from substituting $r_4 = t/x$ into f_4 and clearing denominators.

Proposition 1. *The minimal polynomial of $\rho \cdot r_4$ over $\mathbb{Q}$ is*

$$P(t) = t^{12} - 3t^9 - 2t^8 + 2t^6 - t^5 - 3t^4 - t^3 + t - 1. \tag{2}$$

It is irreducible over $\mathbb{Q}$ and has exactly two real roots: $\rho \cdot r_4 \approx 1.61714$ and ≈ -0.95975.

Proof. The resultant computation yields (2) (verified symbolically in SymPy and confirmed by evaluating $P(\rho \cdot r_4)$ to 50 decimal digits, obtaining a residual of order 10^{-48}). Irreducibility over $\mathbb{Q}$ is confirmed by checking that P has no rational roots (the only candidates ± 1 give $P(1) = -7$ and $P(-1) = -1$) and that P admits no factorization over $\mathbb{Z}$ of degrees 1+11, 2+10, 3+9, 4+8, or 6+6 (verified computationally). □

3 The Fundamental Identity

Theorem 2 (Unit-Norm Remainder)**.** *The polynomial $P(t)$ from (2) satisfies the exact division identity*

$$P(t) = (t^2 - t - 1)\, q(t) + (t - 1), \tag{3}$$

where

$$q(t) = t^{10} + t^9 + 2t^8 + 2t^4 + t^3. \tag{4}$$

Proof. Polynomial long division of (2) by $t^2 - t - 1$ over $\mathbb{Z}[t]$. The computation is routine and can be checked by expanding $(t^2 - t - 1)(t^{10} + t^9 + 2t^8 + 2t^4 + t^3) + (t - 1)$ and verifying that it equals $P(t)$ term by term. □

Since $P(\rho \cdot r_4) = 0$, substituting $t = \rho \cdot r_4$ into (3) yields:

Corollary 3 (Exact Deficit Identity)**.**

$$(\rho\, r_4)^2 - \rho\, r_4 - 1 \;=\; -\,\frac{\rho\, r_4 - 1}{q(\rho\, r_4)}\,, \tag{5}$$

where $q(\rho\, r_4) = 309.4026\ldots$.

The left side of (5) measures how far $\rho \cdot r_4$ deviates from satisfying the golden ratio equation $t^2 = t + 1$. Its magnitude is $|\rho\, r_4 - 1|/q(\rho\, r_4) \approx 0.617/309.4 \approx 0.00199$. The numerator is $O(1)$, but the degree-10 quotient polynomial evaluates to ~ 309 at $\rho \cdot r_4$, suppressing the deficit by more than two orders of magnitude.

By Newton's method, $\varphi - \rho\, r_4 \approx (\rho\, r_4 - 1)/(\sqrt{5}\, q(\rho\, r_4)) = 0.000892\ldots$, in agreement with the directly computed difference $\varphi - \rho\, r_4 = 0.000892378\ldots$ to four significant figures.

4 Algebraic Norm Interpretation

The remainder $t-1$ in (3) has a natural interpretation in the ring of integers of $\mathbb{Q}(\sqrt{5})$.

Definition 4. *For a polynomial $F(t) \in \mathbb{Z}[t]$, define its* golden-ratio residue norm *as*

$$\mathrm{N}_\varphi(F) = \mathrm{N}_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}\big(F(t) \bmod (t^2 - t - 1)\big),$$

where $\mathrm{N}_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}(a\varphi + b) = -a^2 + ab + b^2$ is the field norm.

The remainder $t-1$ represents $\varphi - 1 = 1/\varphi = \varphi^{-1}$ in $\mathbb{Z}[\varphi]$, the ring of integers of $\mathbb{Q}(\sqrt{5})$. Its norm is

$$\mathrm{N}_\varphi(P) = \mathrm{N}(1 \cdot \varphi + (-1)) = -(1)^2 + (1)(-1) + (-1)^2 = -1.$$

This is the norm of the *fundamental unit* of $\mathbb{Q}(\sqrt{5})$. Elements of norm ± 1 in $\mathbb{Z}[\varphi]$ are precisely the units $\pm\varphi^k$ for $k \in \mathbb{Z}$, and the remainder $t - 1 = \varphi^{-1}$ is the simplest such unit.

Theorem 5 (Uniqueness)**.** *Let $P_{m,n}(t)$ be the minimal polynomial of $r_m \cdot r_n$ over $\mathbb{Q}$ for $2 \le m < n$. Then:*

(a) $\mathrm{N}_\varphi(P_{3,4}) = -1$.

(b) $|\mathrm{N}_\varphi(P_{m,n})| \ge 11$ *for all* $(m,n) \ne (3,4)$ *with* $2 \le m < n \le 8$.

(c) $|\mathrm{N}_\varphi(P_{m,n})| \to \infty$ *as* $\max(m,n) \to \infty$.

Proof. Part (a) is Theorem 2. For part (b), we compute $P_{m,n}(t)$ via the resultant $\mathrm{Res}_x(x^m - x - 1,\, t^n - t\,x^{n-1} - x^n)$ for all $\binom{7}{2} = 21$ pairs with $2 \le m < n \le 8$, then reduce modulo $t^2 - t - 1$ and compute the field norm. The results are tabulated in Table 1. The smallest non-unit norm is $|\mathrm{N}_\varphi(P_{2,3})| = 11$.

For part (c), note that $\deg P_{m,n} = m \cdot n$ (since the number fields $\mathbb{Q}(r_m)$ and $\mathbb{Q}(r_n)$ are linearly disjoint over $\mathbb{Q}$ for distinct irreducible trinomials of different degrees). The remainder $R_{m,n}(t) = a_{m,n}\,t + b_{m,n}$ has integer coefficients that are bounded in terms of the coefficients of $P_{m,n}$, and these grow with the resultant, which has entries of magnitude $O(r_m^n + r_n^m)$. Since $r_k \to 1^+$ as $k \to \infty$, the growth is controlled by the polynomial degree $mn \to \infty$, forcing $|a_{m,n}|, |b_{m,n}| \to \infty$ and hence $|\mathrm{N}_\varphi(P_{m,n})| \to \infty$. □

5 Entropy Stacking Corollary

The topological entropy of the substitution dynamical system associated to $x^n = x + 1$ is $h_n = \ln r_n$ [3]. Theorem 2 has a direct consequence for the additivity of these entropies.

Corollary 6 (Entropy Stacking)**.** *Define $h_n = \ln r_n$ for $n \ge 2$. Then*

$$h_2 - (h_3 + h_4) = +0.000552\ldots = 0.115\% \text{ of } h_2,$$

while for all other consecutive triples $(n, n{+}1, n{+}2)$ with $n \ge 3$:

$$h_n - (h_{n+1} + h_{n+2}) < -0.07,$$

a deficit exceeding 26% of h_n.

Table 1: Golden-ratio residue norms $\mathrm{N}_\varphi(P_{m,n})$ for products $r_m \cdot r_n$ from the family $x^k = x+1$.

(m,n)	$r_m \cdot r_n$	$\mathrm{N}_\varphi(P_{m,n})$	Remainder mod (t^2-t-1)
$(3,4)$	$1.61714\ldots$	-1	$t-1$
$(2,3)$	$2.14344\ldots$	11	$-7t-5$
$(2,4)$	$1.97521\ldots$	-29	$-22t-13$
$(2,5)$	$1.88874\ldots$	76	$-54t-34$
$(2,6)$	$1.83602\ldots$	-199	$-145t-89$
$(3,5)$	$1.54635\ldots$	-479	$-121t-73$
$(4,5)$	$1.42498\ldots$	$-7{,}129$	$2987t+1845$
$(3,6)$	$1.50319\ldots$	$-2{,}729$	$805t+496$
$(5,6)$	$1.32457\ldots$	$-2{,}002{,}189$	$508970t+314559$

Proof. By Corollary 3, $(\rho\, r_4)^2 - \rho\, r_4 - 1 = -(\rho\, r_4 - 1)/q(\rho\, r_4)$, so $\rho\, r_4 = \varphi - \varepsilon$ where $\varepsilon = (\rho\, r_4 - 1)/(\sqrt{5}\, q(\rho\, r_4)) + O(\varepsilon^2)$. Then

$$h_2 - (h_3 + h_4) = \ln\varphi - \ln(\rho\, r_4) = \ln\left(\frac{\varphi}{\rho\, r_4}\right) = \frac{\varepsilon}{\varphi} + O(\varepsilon^2) = \frac{\rho\, r_4 - 1}{\varphi\sqrt{5}\, q(\rho\, r_4)} + O(q^{-2}).$$

The denominator $\varphi\sqrt{5}\, q(\rho\, r_4) \approx 1119$ suppresses the $O(1)$ numerator, yielding the claimed surplus of 5.5×10^{-4}.

For $n \geq 3$, the products $r_n \cdot r_{n+1}$ have $|\mathrm{N}_\varphi(P_{n,n+1})| \geq 7{,}129$ (Table 1), and the corresponding deficits are $O(1)$, not $O(q^{-1})$. Direct computation confirms the claimed bounds. □

The golden ratio entropy $h_2 = \ln\varphi$ is thus the unique value in the sequence $\{h_n\}$ that can serve as the "boundary" entropy for the pair (h_3, h_4) in the sense that $h_2 \approx h_3 + h_4$.

6 Remarks

Remark 7. *The Fibonacci connection is visible in the remainder. The pair $(a,b) = (1,-1)$ in the remainder $t-1$ satisfies $a = F_2$, $b = -F_1$ where F_k denotes the k-th Fibonacci number. The Pell equation $-a^2 + ab + b^2 = \pm 1$ over $\mathbb{Z}[\varphi]$ has solutions $(a,b) = \pm(F_{k+1}, -F_k)$ for all $k \geq 1$, but the actual remainder coefficients from the resultant computation do not follow Fibonacci growth for any pair other than $(3,4)$.*

Remark 8. *The resultant $\mathrm{Res}_t(P_{3,4}(t),\, t^2-t-1) = -1$. This equals $P_{3,4}(\varphi) \cdot P_{3,4}(\bar{\varphi})$ where $\bar{\varphi} = (1-\sqrt{5})/2$. The individual values are $P_{3,4}(\varphi) = 1/\varphi$ and $P_{3,4}(\bar{\varphi}) = -\varphi$, whose product is $-1 = 1/(\varphi\bar{\varphi})$, reflecting the unit norm.*

Remark 9. *The quotient $q(t) = t^{10} + t^9 + 2t^8 + 2t^4 + t^3$ has a notable gap structure: the coefficients of $t^7, t^6, t^5, t^2, t^1, t^0$ are all zero. The nonzero terms cluster at the extremes (degrees 8–10 and 3–4), with the high-degree cluster contributing approximately 95% of $q(\rho\, r_4)$. The origin of this sparsity in terms of the Galois structure of the compositum $\mathbb{Q}(r_3, r_4)$ remains unexplored.*

7 Computational Verification

All computations were performed in Python using SymPy (symbolic algebra) and mpmath (arbitrary precision arithmetic at 50 decimal digits). The key results are independently verifiable in three lines:

```
from sympy import Symbol, Poly, resultant
t, x = Symbol('t'), Symbol('x')
P = resultant(Poly(x**3-x-1, x), Poly(t**4-t*x**3-x**4, x), x)
print(divmod(Poly(P, t), Poly(t**2-t-1, t)))
```

Output: `(Poly(t**10 + t**9 + 2*t**8 + 2*t**4 + t**3, t), Poly(t - 1, t))`

Source code reproducing all results is available at `https://github.com/stalex444/golden-ratio-unit-r`

References

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